FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: contingency
Model status: ALL_ENUMERATED_BRANCH_OUTAGES_WITHIN_SUPPLIED_LIMITS
Captured UTC: 2026-10-26T06:04:44.980Z
Warning: ENGINEERING SCREENING ONLY ? independently verify actual manufacturer, utility, installation, e
quipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certif
ication or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: network.baseMVA | Entered value: 10
Input path: network.buses[0].id | Entered value: S
Input path: network.buses[0].type | Entered value: SLACK
Input path: network.buses[0].vSetPu | Entered value: 1
Input path: network.buses[1].id | Entered value: A
Input path: network.buses[1].type | Entered value: PQ
Input path: network.buses[1].pMW | Entered value: -2
Input path: network.buses[1].qMvar | Entered value: -0.7
Input path: network.buses[2].id | Entered value: B
Input path: network.buses[2].type | Entered value: PQ
Input path: network.buses[2].pMW | Entered value: -1.5
Input path: network.buses[2].qMvar | Entered value: -0.5
Input path: network.branches[0].id | Entered value: SA
Input path: network.branches[0].from | Entered value: S
Input path: network.branches[0].to | Entered value: A
Input path: network.branches[0].rPu | Entered value: 0.01
Input path: network.branches[0].xPu | Entered value: 0.08
Input path: network.branches[1].id | Entered value: AB
Input path: network.branches[1].from | Entered value: A
Input path: network.branches[1].to | Entered value: B
Input path: network.branches[1].rPu | Entered value: 0.015
Input path: network.branches[1].xPu | Entered value: 0.09
Input path: network.branches[2].id | Entered value: SB
Input path: network.branches[2].from | Entered value: S
Input path: network.branches[2].to | Entered value: B
Input path: network.branches[2].rPu | Entered value: 0.012
Input path: network.branches[2].xPu | Entered value: 0.1
Input path: nominalBusVoltageV.S | Entered value: 13800
Input path: nominalBusVoltageV.A | Entered value: 13800
Input path: nominalBusVoltageV.B | Entered value: 13800
Input path: busVoltageLimitsPu.S.min | Entered value: 0.9
Input path: busVoltageLimitsPu.S.max | Entered value: 1.1
Input path: busVoltageLimitsPu.A.min | Entered value: 0.9
Input path: busVoltageLimitsPu.A.max | Entered value: 1.1
Input path: busVoltageLimitsPu.B.min | Entered value: 0.9
Input path: busVoltageLimitsPu.B.max | Entered value: 1.1
Input path: branchCurrentLimitsA.SA | Entered value: 300
Input path: branchCurrentLimitsA.AB | Entered value: 300
Input path: branchCurrentLimitsA.SB | Entered value: 300
Input path: maxCases | Entered value: 200

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: ALL_ENUMERATED_BRANCH_OUTAGES_WITHIN_SUPPLIED_LIMITS
Calculated output path: base.outagedBranchId | Value: null

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Calculated output path: base.status | Value: SCREEN_WITHIN_SUPPLIED_LIMITS
Calculated output path: base.violations | Value: []
Calculated output path: base.buses[0].id | Value: S
Calculated output path: base.buses[0].type | Value: SLACK
Calculated output path: base.buses[0].voltagePu | Value: 1
Calculated output path: base.buses[0].angleDegrees | Value: 0
Calculated output path: base.buses[0].pMW | Value: 3.5075898004676587
Calculated output path: base.buses[0].qMvar | Value: 1.2617827725636488
Calculated output path: base.buses[1].id | Value: A
Calculated output path: base.buses[1].type | Value: PQ
Calculated output path: base.buses[1].voltagePu | Value: 0.9924647354031925
Calculated output path: base.buses[1].angleDegrees | Value: -0.8677465652836259
Calculated output path: base.buses[1].pMW | Value: -1.999999999999881
Calculated output path: base.buses[1].qMvar | Value: -0.6999999999998541
Calculated output path: base.buses[2].id | Value: B
Calculated output path: base.buses[2].type | Value: PQ
Calculated output path: base.buses[2].voltagePu | Value: 0.9927454503800234
Calculated output path: base.buses[2].angleDegrees | Value: -0.8506459792576777
Calculated output path: base.buses[2].pMW | Value: -1.4999999999999298
Calculated output path: base.buses[2].qMvar | Value: -0.4999999999999176
Calculated output path: base.branches[0].id | Value: SA
Calculated output path: base.branches[0].from | Value: S
Calculated output path: base.branches[0].to | Value: A
Calculated output path: base.branches[0].currentPu | Value: 0.20918139210024778
Calculated output path: base.branches[0].fromMW | Value: 1.9675659965516556
Calculated output path: base.branches[0].fromMvar | Value: 0.7101897840109007
Calculated output path: base.branches[0].toMW | Value: -1.9631903110715556
Calculated output path: base.branches[0].toMvar | Value: -0.6751843001701028
Calculated output path: base.branches[0].lossMW | Value: 0.004375685480099978
Calculated output path: base.branches[0].lossMvar | Value: 0.03500548384079802
Calculated output path: base.branches[1].id | Value: AB
Calculated output path: base.branches[1].from | Value: A
Calculated output path: base.branches[1].to | Value: B
Calculated output path: base.branches[1].currentPu | Value: 0.004473043531894921
Calculated output path: base.branches[1].fromMW | Value: -0.03680968892832544
Calculated output path: base.branches[1].fromMvar | Value: -0.024815699829767175
Calculated output path: base.branches[1].toMW | Value: 0.03681269014609117
Calculated output path: base.branches[1].toMvar | Value: 0.02483370713636158
Calculated output path: base.branches[1].lossMW | Value: 0.000003001217765731534
Calculated output path: base.branches[1].lossMvar | Value: 0.000018007306594402216
Calculated output path: base.branches[2].id | Value: SB
Calculated output path: base.branches[2].from | Value: S
Calculated output path: base.branches[2].to | Value: B
Calculated output path: base.branches[2].currentPu | Value: 0.16358264399527417
Calculated output path: base.branches[2].fromMW | Value: 1.5400238039160024
Calculated output path: base.branches[2].fromMvar | Value: 0.5515929885527435
Calculated output path: base.branches[2].toMW | Value: -1.5368126901460244
Calculated output path: base.branches[2].toMvar | Value: -0.524833707136259
Calculated output path: base.branches[2].lossMW | Value: 0.003211113769978069
Calculated output path: base.branches[2].lossMvar | Value: 0.02675928141648458
Calculated output path: base.iterations | Value: 3
Calculated output path: base.mismatchPu | Value: 1.4585554986012994e-14
Calculated output path: contingencies[0].outagedBranchId | Value: SA
Calculated output path: contingencies[0].status | Value: SCREEN_WITHIN_SUPPLIED_LIMITS
Calculated output path: contingencies[0].violations | Value: []
Calculated output path: contingencies[0].buses[0].id | Value: S
Calculated output path: contingencies[0].buses[0].type | Value: SLACK
Calculated output path: contingencies[0].buses[0].voltagePu | Value: 1
Calculated output path: contingencies[0].buses[0].angleDegrees | Value: 0
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Calculated output path: contingencies[0].buses[0].pMW | Value: 3.524328032662176
Calculated output path: contingencies[0].buses[0].qMvar | Value: 1.386126454413894
Calculated output path: contingencies[0].buses[1].id | Value: A
Calculated output path: contingencies[0].buses[1].type | Value: PQ
Calculated output path: contingencies[0].buses[1].voltagePu | Value: 0.9727687016520419
Calculated output path: contingencies[0].buses[1].angleDegrees | Value: -2.974874327767519
Calculated output path: contingencies[0].buses[1].pMW | Value: -1.9999999980560488
Calculated output path: contingencies[0].buses[1].qMvar | Value: -0.699999999542174
Calculated output path: contingencies[0].buses[2].id | Value: B
Calculated output path: contingencies[0].buses[2].type | Value: PQ
Calculated output path: contingencies[0].buses[2].voltagePu | Value: 0.9824835672489097
Calculated output path: contingencies[0].buses[2].angleDegrees | Value: -1.9586719194506783
Calculated output path: contingencies[0].buses[2].pMW | Value: -1.5000000014396973
Calculated output path: contingencies[0].buses[2].qMvar | Value: -0.49999999891069036
Calculated output path: contingencies[0].branches[0].id | Value: AB
Calculated output path: contingencies[0].branches[0].from | Value: A
Calculated output path: contingencies[0].branches[0].to | Value: B
Calculated output path: contingencies[0].branches[0].currentPu | Value: 0.21782793838422676
Calculated output path: contingencies[0].branches[0].fromMW | Value: -1.99999999805605
Calculated output path: contingencies[0].branches[0].fromMvar | Value: -0.6999999995421735
Calculated output path: contingencies[0].branches[0].toMW | Value: 2.0071173496671584
Calculated output path: contingencies[0].branches[0].toMvar | Value: 0.7427041092088237
Calculated output path: contingencies[0].branches[0].lossMW | Value: 0.0071173516111081025
Calculated output path: contingencies[0].branches[0].lossMvar | Value: 0.04270410966665014
Calculated output path: contingencies[0].branches[1].id | Value: SB
Calculated output path: contingencies[0].branches[1].from | Value: S
Calculated output path: contingencies[0].branches[1].to | Value: B
Calculated output path: contingencies[0].branches[1].currentPu | Value: 0.3787114287875988
Calculated output path: contingencies[0].branches[1].fromMW | Value: 3.524328032662176
Calculated output path: contingencies[0].branches[1].fromMvar | Value: 1.3861264544138825
Calculated output path: contingencies[0].branches[1].toMW | Value: -3.5071173511068547
Calculated output path: contingencies[0].branches[1].toMvar | Value: -1.242704108119538
Calculated output path: contingencies[0].branches[1].lossMW | Value: 0.01721068155532135
Calculated output path: contingencies[0].branches[1].lossMvar | Value: 0.14342234629434444
Calculated output path: contingencies[0].iterations | Value: 3
Calculated output path: contingencies[0].mismatchPu | Value: 1.9439513887498094e-10
Calculated output path: contingencies[1].outagedBranchId | Value: AB
Calculated output path: contingencies[1].status | Value: SCREEN_WITHIN_SUPPLIED_LIMITS
Calculated output path: contingencies[1].violations | Value: []
Calculated output path: contingencies[1].buses[0].id | Value: S
Calculated output path: contingencies[1].buses[0].type | Value: SLACK
Calculated output path: contingencies[1].buses[0].voltagePu | Value: 1
Calculated output path: contingencies[1].buses[0].angleDegrees | Value: 0
Calculated output path: contingencies[1].buses[0].pMW | Value: 3.5076028298802298
Calculated output path: contingencies[1].buses[0].qMvar | Value: 1.2618366909259215
Calculated output path: contingencies[1].buses[1].id | Value: A
Calculated output path: contingencies[1].buses[1].type | Value: PQ
Calculated output path: contingencies[1].buses[1].voltagePu | Value: 0.9922215257566835
Calculated output path: contingencies[1].buses[1].angleDegrees | Value: -0.88353270655452
Calculated output path: contingencies[1].buses[1].pMW | Value: -1.999999999999863
Calculated output path: contingencies[1].buses[1].qMvar | Value: -0.6999999999997975
Calculated output path: contingencies[1].buses[2].id | Value: B
Calculated output path: contingencies[1].buses[2].type | Value: PQ
Calculated output path: contingencies[1].buses[2].voltagePu | Value: 0.9930472478399631
Calculated output path: contingencies[1].buses[2].angleDegrees | Value: -0.8308649404570259
Calculated output path: contingencies[1].buses[2].pMW | Value: -1.49999999999994
Calculated output path: contingencies[1].buses[2].qMvar | Value: -0.4999999999999116
Calculated output path: contingencies[1].branches[0].id | Value: SA
Calculated output path: contingencies[1].branches[0].from | Value: S

Calculated output path: contingencies[1].branches[0].to | Value: A
Calculated output path: contingencies[1].branches[0].currentPu | Value: 0.21355735136118487
Calculated output path: contingencies[1].branches[0].fromMW | Value: 2.0045606742319033
Calculated output path: contingencies[1].branches[0].fromMvar | Value: 0.7364853938561136
Calculated output path: contingencies[1].branches[0].toMW | Value: -1.9999999999998628
Calculated output path: contingencies[1].branches[0].toMvar | Value: -0.6999999999997899
Calculated output path: contingencies[1].branches[0].lossMW | Value: 0.004560674232040596
Calculated output path: contingencies[1].branches[0].lossMvar | Value: 0.036485393856323656
Calculated output path: contingencies[1].branches[1].id | Value: SB
Calculated output path: contingencies[1].branches[1].from | Value: S
Calculated output path: contingencies[1].branches[1].to | Value: B
Calculated output path: contingencies[1].branches[1].currentPu | Value: 0.15922090651007112
Calculated output path: contingencies[1].branches[1].fromMW | Value: 1.5030421556483275
Calculated output path: contingencies[1].branches[1].fromMvar | Value: 0.5253512970697883
Calculated output path: contingencies[1].branches[1].toMW | Value: -1.499999999999941
Calculated output path: contingencies[1].branches[1].toMvar | Value: -0.4999999999998995
Calculated output path: contingencies[1].branches[1].lossMW | Value: 0.003042155648386724
Calculated output path: contingencies[1].branches[1].lossMvar | Value: 0.025351297069888812
Calculated output path: contingencies[1].iterations | Value: 3
Calculated output path: contingencies[1].mismatchPu | Value: 2.0247692411601292e-14
Calculated output path: contingencies[2].outagedBranchId | Value: SB
Calculated output path: contingencies[2].status | Value: SCREEN_WITHIN_SUPPLIED_LIMITS
Calculated output path: contingencies[2].violations | Value: []
Calculated output path: contingencies[2].buses[0].id | Value: S
Calculated output path: contingencies[2].buses[0].type | Value: SLACK
Calculated output path: contingencies[2].buses[0].voltagePu | Value: 1
Calculated output path: contingencies[2].buses[0].angleDegrees | Value: 0
Calculated output path: contingencies[2].buses[0].pMW | Value: 3.5180750520200554
Calculated output path: contingencies[2].buses[0].qMvar | Value: 1.3367779671252755
Calculated output path: contingencies[2].buses[1].id | Value: A
Calculated output path: contingencies[2].buses[1].type | Value: PQ
Calculated output path: contingencies[2].buses[1].voltagePu | Value: 0.9861521440443572
Calculated output path: contingencies[2].buses[1].angleDegrees | Value: -1.5577356226424277
Calculated output path: contingencies[2].buses[1].pMW | Value: -2.0000000001330154
Calculated output path: contingencies[2].buses[1].qMvar | Value: -0.699999999872444
Calculated output path: contingencies[2].buses[2].id | Value: B
Calculated output path: contingencies[2].buses[2].type | Value: PQ
Calculated output path: contingencies[2].buses[2].voltagePu | Value: 0.9791725987971484
Calculated output path: contingencies[2].buses[2].angleDegrees | Value: -2.314293683063583
Calculated output path: contingencies[2].buses[2].pMW | Value: -1.4999999998054299
Calculated output path: contingencies[2].buses[2].qMvar | Value: -0.49999999995306804
Calculated output path: contingencies[2].branches[0].id | Value: SA
Calculated output path: contingencies[2].branches[0].from | Value: S
Calculated output path: contingencies[2].branches[0].to | Value: A
Calculated output path: contingencies[2].branches[0].currentPu | Value: 0.3763486070790953
Calculated output path: contingencies[2].branches[0].fromMW | Value: 3.518075052020057
Calculated output path: contingencies[2].branches[0].fromMvar | Value: 1.336777967125281
Calculated output path: contingencies[2].branches[0].toMW | Value: -3.5039112246150195
Calculated output path: contingencies[2].branches[0].toMvar | Value: -1.2234673478849807
Calculated output path: contingencies[2].branches[0].lossMW | Value: 0.014163827405037632
Calculated output path: contingencies[2].branches[0].lossMvar | Value: 0.113310619240300 36
Calculated output path: contingencies[2].branches[1].id | Value: AB
Calculated output path: contingencies[2].branches[1].from | Value: A
Calculated output path: contingencies[2].branches[1].to | Value: B
Calculated output path: contingencies[2].branches[1].currentPu | Value: 0.1614770298747223
Calculated output path: contingencies[2].branches[1].fromMW | Value: 1.503911224482005
Calculated output path: contingencies[2].branches[1].fromMvar | Value: 0.5234673480125216
Calculated output path: contingencies[2].branches[1].toMW | Value: -1.4999999998054305
Calculated output path: contingencies[2].branches[1].toMvar | Value: -0.4999999999530758

Calculated output path: contingencies[2].branches[1].lossMW | Value: 0.003911224676574343
Calculated output path: contingencies[2].branches[1].lossMvar | Value: 0.02346734805944571
Calculated output path: contingencies[2].iterations | Value: 3
Calculated output path: contingencies[2].mismatchPu | Value: 1.945701932903887e-11
Calculated output path: incompleteEnumeration | Value: false
Calculated output path: method | Value: Balanced AC Newton-Raphson, every enumerated single branch outage; explicit voltage/current limits; not IEEE/IEC equipment compliance or dynamic stability

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY ? independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: contingency
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:44.980Z